MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCMBG101 – Wonders of the Microbial World

Duration: 1½ hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A
Multiple Choice Questions
Answer **All** Questions
Each question carries **1** mark

1. Who developed vaccination against rabies? [K] [1]
   a) Louis Pasteur b) Robert Koch c) Edward Jenner d) Paul Ehrlich
2. Which part of the microscope holds the objective lenses? [K] [1]
   a) Body tube b) Nose piece c) Condenser d) Stage
3. Select the primary stain used in endospore staining. [K] [2]
   a) Crystal violet b) Carbol fuchsin c) Methylene blue d) Malachite green
4. The protein coat of virus is called ..... [U] [3]
   a) nucleoid b) viroid c) capsid d) plasmid
5. Select the target site of penicillin in the bacterial cell. [A] [4]
   a) Cell wall b) cell membrane c) DNA synthesis d) protein synthesis
6. Who among the following is not a supporter of spontaneous generation theory? [K] [1]
   a) Aristotle b) Epicuris c) Louis Pasteur d) Needham
7. Which of the following microscopes is the best suited for observing *Treponema pallidum*?]K] [2]
   a) Fluorescence microscope b) Bright field microscope
   c) Phase contrast microscope d) Dark field microscope
8. Select the magnification power of oil immersion objective. [A] [2]
   a) 10 X b) 40 X c) 100 X d) 1000 X

9. What are prions? [K] [3]

a) plasmid proteins b) infectious proteins c) virus proteins d) fungal proteins

10. Select the temperature-time used in autoclave sterilization. [A] [4]

a. 72 C, 15 sec b. 170 C, 1 hr c. 121 C, 15 min d. 100 C, 20 min

[1 x 10 = 10]

Part B

Fill in the blanks

Answer **All** Questions

Each question carries **1** mark

11. ..... is an example of a basic stain. [K] [2]

12. Fungi that have no known sexual state in their life cycle are called ..... [K] [3]

13. The locomotor organ in bacteria is ..... [U] [3]

14. ..... is known as the Father of antiseptic surgery. [K] [1]

15. The lowest concentration of an antibiotic that can inhibit the growth of microorganisms is called ..... [U] [4]

[1 x 5 = 5]

Part C

State whether True or False

Answer **All** Questions

Each question carries **1** mark

16. The extra-chromosomal DNA in bacteria are called plasmids. [U] [3]

17. The primary stain in Gram stain is safranin. [U] [2]

18. Salvarsan is the first magic bullet developed against syphilis bacteria. [K] [1]

19. Erythromycin inhibits protein synthesis in bacteria. [U] [4]

20. Hot air oven works on the principle of moist heat. [U] [4]

[1 x 5 = 5]

Part D

Very Short Answer Type Questions

Answer **10** Questions

Each question carries **1** mark

21. What are viroids? [U] [3]

22. Define prokaryotes. [K] [3]

23. Explain the use of biofertilizers. [U] [1]
24. Explain theory of spontaneous generation. [U] [1]
25. What role does an exciter filter play in a fluorescence microscope? [K] [2]

26. Explain resolution of a microscope. [U] [2]
27. Compare microbicidal and microbistatic agents. [U] [4]
28. Define sterilization. [K] [4]
29. Explain wet mount. [U] [2]
30. Explain pasteurization method. [U] [4]
31. Who is known as the father of antibiotics? [K] [1]
32. What are bacteriophages? [K] [3]

[1 x 10 = 10]

Part E
Short Answer Type Questions
Answer **4** Questions
Each question carries **3** mark

33. Summarize the contributions of Louis Pasteur in developing Microbiology as an applied branch of science. [U] [1]
34. Find out the role of microorganisms in improvement of soil fertility. [K] [1]
35. Explain the working of dark field microscope. [U] [2]
36. Outline the steps in lytic cycle. Explain each step. [U] [3]
37. Explain disc diffusion method. [U] [4]
38. Explain the principle and working of an autoclave. [U] [4]

[3 x 4 = 12]

Part F
Long Answer Type Questions
Answer **1** Question
The question carries **8** marks

39. Analyze the ultra structure of a bacterial cell. [An] [3]
40. Outline the principle and organize the steps in the procedure of Gram's staining. [U] [2]

[1 x 8 = 8]